



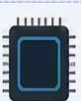
CHAPTER III : SOFTWARE



Software is a set of instructions, data, or programs used to operate computers and perform specific tasks. It is broadly categorized into three main types.

1. SYSTEM SOFTWARE

System software manages computer hardware and provides a platform for running application software.

-  **Operating Systems (OS):** The primary software that boots a computer and manages resources like memory, CPU, and storage.
-  **Device Drivers:** Programs that allow the OS to communicate with hardware devices such as printers, scanners, and graphics cards.
-  **Language Processors:** Convert high-level programming languages (like Java, C++, Python) into machine-readable code.
-  **Firmware:** Embedded software that controls hardware functions, often found in devices like routers or cameras.

2. APPLICATION SOFTWARE

Application software is designed for end-users to perform specific tasks.

-  **Desktop Applications:** Programs installed on a computer, such as Microsoft Word, Excel, or Adobe Photoshop.
-  **Mobile Applications:** Apps for smartphones and tablets, like Instagram, WhatsApp, or MyFitnessPal.
-  **Web Applications:** Software accessed via web browsers, such as Google Docs or online banking platforms.
-  **Enterprise Software:** Supports large organizations with operations like accounting, CRM, or ERP.
-  **Cloud-Based Software (SaaS):** Accessible over the internet, offering collaboration and scalability without local installation.

3. SPECIALIZED SOFTWARE

Specialized software serves specific industries or advanced functions.

-  **Programming Software:** Tools like compilers, interpreters, and IDEs that help developers write and test code.
-  **Database Management Systems (DBMS):** Software for storing, managing, and retrieving data, such as MySQL or Oracle.
-  **Embedded Systems:** Software integrated into hardware devices like washing machines, medical devices, or automotive systems.
-  **Artificial Intelligence (AI) Software:** Programs that perform tasks requiring human-like intelligence, including machine learning, NLP, and analytics.
-  **Simulation Software:** Creates virtual models for training or experimentation, used in engineering, aviation, and healthcare.

IMPORTANCE OF SOFTWARE



Makes hardware useful and functional



Improves productivity and efficiency



Supports communication and collaboration



Ensures security and data management



Drives innovation and economic growth

General Objective

To ensure that operating systems and application software are regularly updated, secure, efficient, and compatible with current hardware and software requirements.

Specific Objectives

- To improve system security** by installing updates and patches that fix vulnerabilities and protect against malware and cyber threats.
- To enhance software performance** by applying updates that optimize speed, stability, and overall functionality.
- To maintain compatibility** between operating systems, applications, and hardware devices through regular software updates.
- To fix bugs and errors** that may affect the reliability and usability of computer systems and applications.
- To provide access to new features and improvements** introduced by software developers through updated versions.
- To reduce system downtime and operational issues** by ensuring that software remains current and properly maintained.
- To improve user productivity** by keeping applications running efficiently and minimizing technical problems.

Distinct Objective

To maintain the security, performance, reliability, and compatibility of computer systems through the regular updating of operating systems and application software.

1. SYSTEM SOFTWARE



System software manages computer hardware and provides a platform for running application software.

It includes:

1



OPERATING SYSTEMS (OS):

The primary software that boots a computer and manages resources like memory, CPU, and storage.

Examples:



Windows



macOS



Linux



Android

2



DEVICE DRIVERS:

Programs that allow the OS to communicate with hardware devices such as printers, scanners, and graphics cards.



3

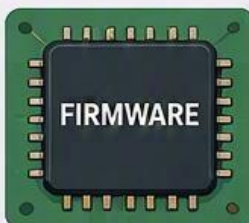


LANGUAGE PROCESSORS:

Convert high-level programming languages (like Java, C++, Python) into machine-readable code.



4



FIRMWARE:

Embedded software that controls hardware functions, often found in devices like routers or cameras.



ROLE OF SYSTEM SOFTWARE



Manages
Hardware
Resources



Provides a Platform
for Application
Software



Ensures System
Security and
Stability



Optimizes
Performance



Coordinates all
System
Operations

2. APPLICATION SOFTWARE

Application software is designed for end-users to perform specific tasks.



COMMON TYPES INCLUDE

1



DESKTOP APPLICATIONS:

Programs installed on a computer, such as Microsoft Word, Excel, or Adobe Photoshop.



Word



Excel



Photoshop

2



MOBILE APPLICATIONS:

Apps for smartphones and tablets, like Instagram, WhatsApp, or MyFitnessPal.



Instagram

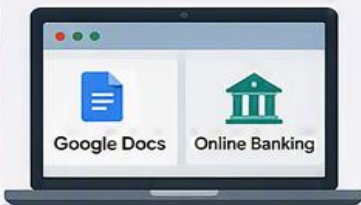


WhatsApp



MyFitnessPal

3



WEB APPLICATIONS:

Software accessed via web browsers, such as Google Docs or online banking platforms.



Google Docs



Online Banking

4



ENTERPRISE SOFTWARE:

Supports large organizations with operations like accounting, customer relationship management (CRM), or resource planning (ERP).



CRM



ERP

5



CLOUD-BASED SOFTWARE (SAAS):

Accessible over the internet, offering collaboration and scalability without local installation.



Google Workspace



Microsoft 365



Slack

BENEFITS OF APPLICATION SOFTWARE



Helps users perform specific tasks efficiently



Improves productivity and accuracy



Facilitates communication and collaboration



Supports business growth and decision-making



Enables access anytime, anywhere (especially SaaS)

3. SPECIALIZED SOFTWARE



Specialized software serves specific industries or advanced functions.

1

PROGRAMMING SOFTWARE:

Tools like compilers, interpreters, and IDEs that help developers write and test code.



VS Code



IntelliJ IDEA



Code::Blocks

2

DATABASE MANAGEMENT SYSTEMS (DBMS):

Software for storing, managing, and retrieving data, such as MySQL or Oracle.



ORACLE

Oracle

3

EMBEDDED SYSTEMS:

Software integrated into hardware devices like washing machines, medical devices, or automotive systems.



Washing Machine



Medical Device



Automotive System

4

ARTIFICIAL INTELLIGENCE (AI) SOFTWARE:

Programs that perform tasks requiring human-like intelligence, including machine learning, natural language processing, and predictive analytics.



Machine Learning



Natural Language Processing



Predictive Analytics



AI Applications

5

SIMULATION SOFTWARE:

Creates virtual models for training or experimentation, used in engineering, aviation, and healthcare.



Engineering Simulation



Aviation Simulation



Healthcare Simulation

USES OF SPECIALIZED SOFTWARE



Solves complex problems



Improves accuracy and efficiency



Supports industry-specific needs



Drives innovation and research



Enhances safety and reliability